



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade

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22CH203 ENGINEERING CHEMISTRY-II

UNIT I

LESSON PLAN 2-Topic 4

SECONDARY REFERENCE ELECTRODE

1. Calomel Electrode or Mercury- Mercurous Chloride Electrode (Secondary Reference Electrode)

The potential of the calomel electrode, on the hydrogen scale, is found to vary with the concentration of the KCl solution. The reduction potentials for the various KCl concentration at 25 °C are,

KCl concentration	0.1N	1.0N	Saturated
Electrode potential (in volt)	0.3335	0.2810	0.2422

Construction

- The calomel electrode is the mercury-mercurous chloride electrode.
- It consists of a glass vessel having a bent side tube.
- Pure mercury is placed at the bottom of the tube, which is covered with a paste of mercury- mercurous chloride ($\text{Hg} + \text{Hg}_2\text{Cl}_2$) that is calomel.
- The remaining portion of the cell is filled with a solution of normal (1 N) or saturated KCl (Potassium chloride).
- A platinum wire sealed into a glass tube is dipped into a mercury layer and is used to provide external electrical contact.
- The side tube is used for making electrical contact with a salt bridge.

- The calomel electrode can act as an anode or cathode depending on the nature of another electrode of the cell.

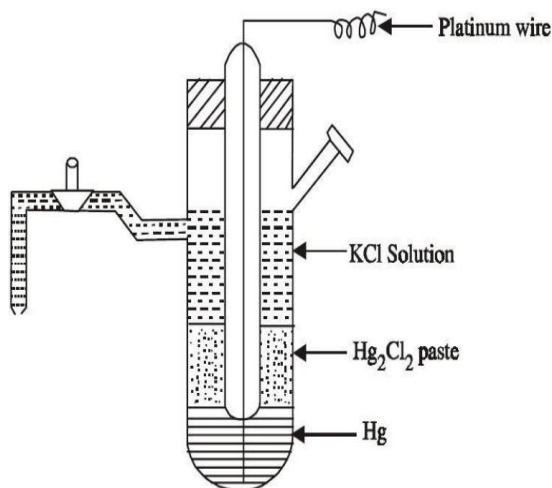
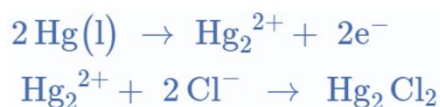


Figure.1 Calomel Electrode

Cell Representation of Saturated Calomel Electrode:

Hg, Hg₂Cl₂(s), KCl (satd. Solution)

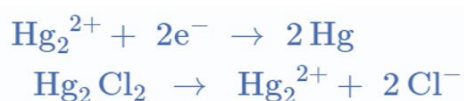
Reaction at the anode:



Overall reaction:



Reaction at the cathode:

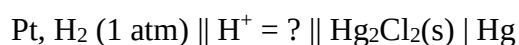


Overall reaction:



Example:

The calomel electrode can be coupled with *hydrogen electrode* containing solution of unknown pH.



The e.m.f. of the cell,

$$E_{\text{cell}} = E_{\text{Right}} - E_{\text{Left}}$$



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$$= 0.2422 \text{ V} + 0.0592 \text{ V pH}$$

$$\text{pH} = \frac{E_{\text{cell}} - 0.2422 \text{ V}}{0.0592}$$

Applications of calomel electrode

- It is used to determine the potentials of other electrodes by developing a reversible galvanic cell by combining with experimental electrode.

Advantages

- It has simple construction
- Cell potential is constant over a long period.
- Cell potential does not vary with temperature
- Cell potential is reproducible

2. Silver/Silver Chloride Reference Electrode

The silver/silver chloride reference electrode is a widely used reference electrode because it is simple, inexpensive, very stable and non-toxic.

Construction

The electrode is constructed using a silver wire, electroplated with silver chloride and then immersed into a 3.5 M or saturated potassium chloride (KCl) solution. The electrolyte is generally contained in a glass tube having a porous glass frit at the tip to allow ion exchange at the liquid-liquid junction.

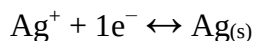
Working

The silver-silver chloride reference electrode develops a potential proportional to the chloride concentration, whether it is NaCl, KCl, NH₄Cl or some other chloride salt and remains constant as long as the chloride concentration remains constant.

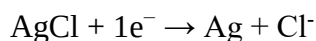
The electrode is represented as:



The cell reaction is



Net reaction



Advantages

- Stable potential.
- Thermal stability
- Non-toxic components.
- Simple construction.
- Inexpensive to manufacture.

Disadvantages

- Short lifetime due to the small amount of Ag/AgCl.
- Its potential is very sensitive to the testing solution due to the lack of a filling solution.
- Application as reference electrodes is limited.

Discussion Questions

1. Compare and contrast the Calomel Electrode with the Standard Hydrogen Electrode (SHE) in terms of construction, operation, and applications.
2. Discuss the advantages and disadvantages of using the Calomel Electrode as a reference electrode compared to other types of reference electrodes.
3. How does the Calomel Electrode contribute to the understanding of electrochemical processes and the measurement of electrode potentials?
4. Explore modern advancements and innovations in reference electrode technology and their impact on the relevance and usage of Calomel Electrodes.
5. Discuss any potential issues or limitations associated with the use of Calomel Electrodes in practical applications.

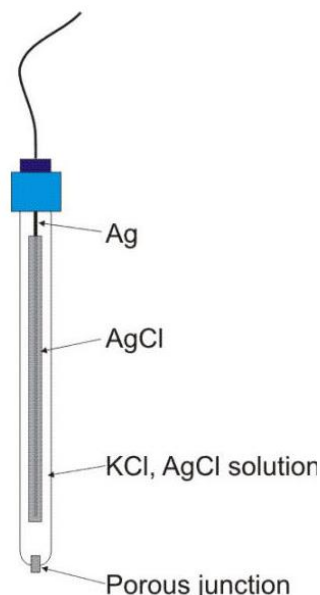


Figure 2 Silver/Silver Chloride Electrode